

Mani K Chettri

PhD Candidate in Space Plasma Physics

Department of Physics, Sikkim University, Gangtok, Sikkim, India

Expected PhD completion: Q1 2027

Available for postdoctoral appointments: from early 2027

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Research Profile

Final-year PhD researcher studying how turbulent energy reaches kinetic scales and is dissipated in collisionless space plasmas. My work combines high-resolution spacecraft observations with pseudospectral simulations of nonlinear plasma models, with emphasis on kinetic Alfvén waves in Earth's magnetosphere and the solar wind. Research interests include kinetic-scale turbulence, nonlinear wave coupling, collisionless dissipation, wave-particle interactions, and plasma instabilities.

Education

PhD in Physics, Sikkim University, Gangtok, India

2022 – present

Expected completion: Q1 2027

Thesis: Numerical Study of Nonlinear Plasma Phenomena in the Heliosphere

Supervisors: Dr. Hemam Dinesh Singh (NSUT, New Delhi) and Dr. Rupak Mukherjee (Sikkim University; IUCAA Visiting Associate)

Focus: KAW turbulence in the solar wind and magnetosphere using modified Zakharov and NLSE models, pseudospectral simulations, and multi-mission spacecraft observations.

MSc in Physics, Sikkim Manipal Institute of Technology, Majitar

2020 – 2022

Dissertation: The Cosmic Microwave Background Radiation and the Big Bang

BSc Physics (Honours), Nar Bahadur Bhandari Degree College, Tadong

2016 – 2019

Publications

Peer-Reviewed Journal Articles

1. **Chettri, M. K.**, Mukherjee, R., and Singh, H. D. (2026). MMS observations of kinetic Alfvén wave turbulence and steep kinetic-range spectra in the outer plasma sheet boundary layer. *Journal of Atmospheric and Solar-Terrestrial Physics*, Article 106920. [DOI](#) [Publisher](#) [arXiv](#)
2. **Chettri, M. K.**, Shrivastav, V., Singh, B., Mukherjee, R., and Singh, H. D. (2026). Inertial-range suppression and ponderomotive density cavitation in broadband sub-Alfvénic turbulence under plasma sheet boundary layer conditions. *Journal of Subatomic Particles and Cosmology*, 100383. [DOI](#) [arXiv](#)
3. Shrivastav, V., **Chettri, M. K.**, Singh, H. D., Singh, B., and Mukherjee, R. (2026). Suppression of two-stream and Weibel instabilities in ultra-relativistic electron-positron plasma. *Journal of Subatomic Particles and Cosmology*, 100421. [DOI](#) [Publisher](#)
4. Shrivastav, V., **Chettri, M. K.**, Singh, H. D., Singh, B., and Mukherjee, R. (2026). Weibel instability in collisionless plasmas across astrophysical and laboratory shocks. *Contributions to Plasma Physics*, Early View. [DOI](#)
5. Mukherjee, R., **Chettri, M. K.**, Shrivastav, V., Singh, B., and Singh, H. D. (2026). How big is the Sun? A measurement from India's Aditya-L1 spacecraft. *PANE Journal of Physics Education*, 1(1), 60–74. [Article](#)
6. **Chettri, M. K.**, Shrivastav, V., Mukherjee, R., Gaur, N., Sharma, R., and Singh, H. D. (2024). Nonlinear coupling of kinetic Alfvén waves and ion acoustic waves in the inner heliosphere. *Research in Astronomy and Astrophysics*, 24(10), 105009. [DOI](#)

Preprints and Submitted Work

1. **Chettri, M. K.**, Mukherjee, R., and Singh, H. D. (2026). Electromagnetic signatures of kinetic Alfvén wave turbulence at ion inertial scales in Earth’s high- β magnetosheath. Submitted to *Advances in Space Research*. [arXiv:2605.25199](#)
2. **Chettri, M. K.**, Shrivastav, V., Singh, B., Mukherjee, R., and Singh, H. D. (2026). A validated analysis workflow for near-Earth spacecraft plasma observations: methodology, quality controls, and systematic error assessment. [SSRN 6394663](#)
3. **Chettri, M. K.**, Singh, H. D., Shrivastav, V., Singh, B., and Mukherjee, R. (2025). Damped kinetic Alfvén waves in Earth’s magnetosheath: numerical simulations and MMS observations. [arXiv:2512.09828](#)
4. Shrivastav, V., **Chettri, M. K.**, Singh, H. D., Singh, B., and Mukherjee, R. (2026). Ion channel dynamics in temperature-dependent Weibel instability saturation. [arXiv:2604.21875](#)
5. Shrivastav, V., **Chettri, M. K.**, Singh, H. D., and Mukherjee, R. (2025). Suppression of two-stream instability in relativistic electron–ion plasmas. [arXiv:2508.14362](#)

Research Software

PLASMAform: Interactive Space Plasma Formulary

2025 – present

[Web application](#) [GitHub](#) MIT License

Browser-based research tool covering 23 formulas in 6 categories, including KAW diagnostics, instability thresholds, spacecraft parameter presets, validation tables, and a worked MMS example.

Conference Presentations

52nd EPS Conference on Plasma Physics (EPS-2026), poster

29 Jun – 3 Jul 2026

Edinburgh International Conference Centre, Edinburgh, UK

Ponderomotive generation of density cavities from kinetic Alfvén wave–slow-mode coupling.

National Conference on Trends in Science and Technology (TIST-2026), contributed talk

27 – 28 Feb 2026

Salesian College, Siliguri, India

Damped kinetic Alfvén waves in Earth’s magnetosheath: numerical simulations and MMS observations.

5th Conference on Plasma Simulation (CPS-2026), poster

Jan 2026

IPS Academy and IIT Indore, India

KAW turbulence simulations in the plasma sheet boundary layer with MMS comparison.

XV PANE Conference 2025, contributed talk

Dec 2025

Nagaland University, Lumami, India

Turbulent signatures of kinetic Alfvén waves in Earth’s plasma sheet boundary layer.

Selected Workshops and Professional Development

13th Aditya-L1 Support Cell Workshop

6 – 15 Jul 2026

ARIES, Nainital, India. Hands-on training with Aditya-L1 data products and analysis.

JPP Frontiers of Plasma Physics Colloquium

May – Jun 2026

Journal of Plasma Physics, Cambridge University Press; online colloquium series on kinetic plasma modelling and machine-learning closure relations.

Aditya-L1 Support Cell Workshop

Dec 2025

IIT Jammu, organised by ISRO. Training with ASPEX, PAPA, and MAG data through ISSDC/PRADAN.

Technical Skills

Programming	Python, Fortran, MATLAB, JavaScript/HTML, $\text{\LaTeX}$
Scientific libraries	NumPy, SciPy, Matplotlib, Astropy, PySPEDAS, mpi4py, mpi4py-fft
Numerical methods	Fourier pseudospectral methods, 2/3 dealiasing, split-step Fourier, fourth-order Runge–Kutta, MPI-parallelised solvers
Spacecraft data	MMS (FGM, SCM, FPI, EDP); Wind (MFI, SWE); Parker Solar Probe; Aditya-L1 (SUIT and SoLEXS analysis; workshop training with ASPEX, PAPA, and MAG)
HPC and tools	MPI, IUCAA HPC cluster, Git, GitHub
Languages	Nepali, Hindi, and English (read, speak, and write)

Teaching Experience

Part-time Science and Mathematics Instructor, Ascend Coaching Institute, Gangtok	2022 – present
Science Teacher, St. Mary’s Convent School, West Sikkim	2019 – 2020
Mathematics Tutor, Grace Academy, Gangtok	2017 – 2019

Professional Memberships

American Physical Society, Division of Plasma Physics	2026 – 2027
European Geosciences Union, Solar-Terrestrial Sciences Division	2026
European Physical Society, Individual Member	2026

References

Dr. Hemam Dinesh Singh

Associate Professor
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Dr. Rupak Mukherjee

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